Karlsruhe Institute of Technology

Test, Validation, Analysis

Title of the Thesis

Bachelor's Thesis
of

Name Surname

at the Department of Informatics
Institute of Information Security and Dependability
Test, Validation and Analysis (TVA)

Reviewer:	Prof. Dr.
Second Reviewer:	Prof. Dr.
Advisors:	M.Sc.
	Dr.

Completion period: ??. Month 2023 – ??. Month 2023

Introduction

The introduction is the main part of your proposal where you describe the motivation for you topic and how you will address it. Start by describing a bigger problem in industry or research, to set the locate your thesis into this research area. Then breakdown the big topic into smaller topics and existing research areas. Focus especially on the research area your thesis will be placed in. Describe the state of the art in this research are and explain why existing approaches do not solve the challenges mentioned before. Now name the topic of your thesis and the overall research goal you want to analyse with it. Then split the overall research question into smaller research question that help you to achieve your overall research goal. Describe each research question by defining tasks you need to perform to answer the research question and by describing how you can evaluate the quality of the results from those tasks.

. . .

Evaluation

In this section you elaborate on how you will evaluate the results of your thesis. Try to think about metrics and evaluation procedure you need to proof the quality of your defined research tasks. Describe the concept including the tasks you need to perform. Describe which subject systems you have in mind to perform your evaluation. Reason about why you use those systems and define what you have to change or prepare in advance to make those subject systems usable for you.

. . .

Workplan

In this section you will define work packages for your thesis and define a time schedule to work on them. Start by thinking about which tasks you need to perform to answer your research questions from the introduction. Define headlines for your tasks, split bigger tasks into sub-tasks and think about how critical those sub-tasks are for the success of your thesis (must, should, can) Then describe in detail, what you have to do, in those tasks and sub-tasks. After you have your work packages defined, make a time table to structure your work over the time of your thesis. Remember to allocate time for writing the thesis and work with a buffer in case of unforeseen events. Visualize your time schedule. . . .

A bachelor thesis at KIT goes for four months (16 weeks). Remove bachelor time schedule if you are doing a masters thesis

A masters thesis at KIT goes for six months (24 weeks). Remove master time schedule if you are doing a bachelor thesis

Work Packages	Weeks															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Work Package 1																
Work Package 2																
Work Package 3																
Buffer																
Writing																
Presentations																◆

Table 1: Example time schedule for bachelor thesis

Work Packages	Weeks																							
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Work Package 1																								
Work Package 2																								
Work Package 3																								
Buffer																								
Writing																								
Presentations												◆												◆

Table 2: Example time schedule for master thesis

References

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[4] Kyo C. Kang, Sholom G. Cohen, James A. Hess, William E. Novak, and A. Spencer Peterson. Feature-Oriented Domain Analysis (FODA) Feasibility Study. Technical Report CMU/SEI-90-TR-21, Software Engineering Institute, 1990.

[5] Cory J. Kapser and Michael W. Godfrey. Supporting the Analysis of Clones in Software Systems: A Case Study. *Journal of Software Maintenance and Evolution*, 18(2):61–82, 2006.

[6] Christian Kästner, Sven Apel, and Martin Kuhlemann. A Model of Refactoring Physically and Virtually Separated Features. In *Proceedings of the International Conference on Generative Programming and Component Engineering (GPCE)*, pages 157–166. ACM, October 2009.

[7] Malte Rosenthal. Alternative Features in Colored Featherweight Java. Diplomarbeit, University of Passau, Germany, July 2009.